WORKED SOLUTION

STEP II 2009, Question 12

Probability and Statistics • Half-Normal Distribution

STEP II 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q12

The key idea

The exponent $-2x^2$ is a disguised normal density. Writing $-2x^2 = -\dfrac{x^2}{2\sigma^2}$ gives $\sigma = \frac{1}{2}$, so f is exactly twice the density of $Y \sim \mathrm{N}(0, \frac{1}{4})$, cut off at 0. In other words, X has the same distribution as $|Y|$. Once you spot that, the normalising constant, $\mathrm{E}(X^2)$ and the median all come straight from facts about the normal distribution. Only $\mathrm{E}(X)$ needs a (very easy) integral.

Solution

Throughout, let $Y \sim \mathrm{N}(0, \frac{1}{4})$ and $Z \sim \mathrm{N}(0,1)$, so that Y has the same distribution as $\frac{1}{2}Z$ and its density is

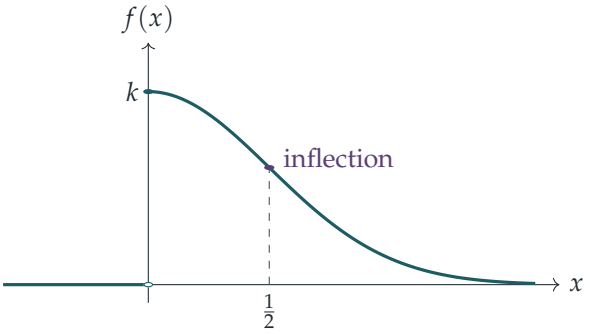
$$g(y) = \frac{1}{\frac{1}{2}\sqrt{2\pi}}\,\mathrm{e}^{-y^2/(2\cdot\frac{1}{4})} = \frac{2}{\sqrt{2\pi}}\,\mathrm{e}^{-2y^2}.$$

Part (i)

For $x \geq 0$, $f'(x) = -4kx\,\mathrm{e}^{-2x^2} \leq 0$, so f decreases from its maximum value k at $x = 0$, with zero gradient there. Also

$$f''(x) = k(16x^2 - 4)\,\mathrm{e}^{-2x^2},$$

which changes sign at $x = \frac{1}{2}$, so there is a point of inflection at $x = \frac{1}{2}$. As $x \to \infty$, $f(x) \to 0$ very rapidly, and $f(x) = 0$ for $x < 0$. The graph is the right-hand half of a bell curve (with $k \approx 1.60$ from part (ii)), and there is a jump at $x = 0$ from 0 up to k.



Part (ii)

The total probability must be 1. Since g is symmetric about 0 and integrates to 1 over the whole real line, its integral over $[0,\infty)$ is $\frac{1}{2}$:

$$\int_0^\infty \frac{2}{\sqrt{2\pi}}\,\mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{2} \quad\Longrightarrow\quad \int_0^\infty \mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{\sqrt{2\pi}}{4}.$$

So $k\cdot\dfrac{\sqrt{2\pi}}{4} = 1$, giving

$$k = \frac{4}{\sqrt{2\pi}} = 2\sqrt{\frac{2}{\pi}} \approx 1.596.$$

Notice that $f(x) = 2g(x)$ for $x \geq 0$, which confirms that X is distributed like $|Y|$.

Part (iii)

For the mean, the factor x makes the integral elementary, because $x\,\mathrm{e}^{-2x^2}$ is a multiple of the derivative of e^{-2x^2}:

$$\mathrm{E}(X) = k\int_0^\infty x\,\mathrm{e}^{-2x^2}\,\mathrm{d}x = k\Big[-\tfrac{1}{4}\mathrm{e}^{-2x^2}\Big]_0^\infty = \frac{k}{4} = \frac{1}{\sqrt{2\pi}} \approx 0.399.$$

? **Ask yourself** How do we find $\displaystyle\int_0^\infty x^2\,\mathrm{e}^{-2x^2}\,\mathrm{d}x$ without a long integration by parts?

Use the normal distribution again. The function $x^2 g(x)$ is even, so

$$\mathrm{E}(X^2) = \int_0^\infty x^2\cdot 2g(x)\,\mathrm{d}x = \int_{-\infty}^\infty x^2 g(x)\,\mathrm{d}x = \mathrm{E}(Y^2) = \mathrm{Var}(Y) = \frac{1}{4},$$

since $\mathrm{E}(Y) = 0$. (Integration by parts with $u = x$, $\mathrm{d}v = x\mathrm{e}^{-2x^2}\,\mathrm{d}x$ gives the same: $\int_0^\infty x^2\mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{4}\int_0^\infty \mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{4k}$.) Hence

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = \frac{1}{4} - \frac{1}{2\pi} = \frac{\pi-2}{4\pi} \approx 0.0908.$$

Part (iv)

The median m satisfies $\mathrm{P}(X \leq m) = \frac{1}{2}$. Since X is distributed like $|Y|$ and Y like $\frac{1}{2}Z$,

$$\mathrm{P}(X \leq m) = \mathrm{P}(|Z| \leq 2m) = 2\Phi(2m) - 1,$$

using the symmetry of the standard normal distribution. (Directly: $\mathrm{P}(X \leq m) = \int_0^m 2g(x)\,\mathrm{d}x = 2(\mathrm{P}(Y \leq m) - \frac{1}{2})$.) Setting this equal to $\frac{1}{2}$ gives

$$\Phi(2m) = \frac{3}{4}.$$

From tables, $\Phi^{-1}(0.75) = 0.6745$, so $2m = 0.6745$ and

$$m \approx 0.337 \quad \text{(to 3 s.f.)}.$$

Checks

A numerical (Simpson's rule) integration of f over $[0, 10]$ with $k = 2\sqrt{2/\pi}$ gives total probability 1.0000, $\mathrm{E}(X) = 0.39894$ and $\mathrm{Var}(X) = 0.090845$, matching $1/\sqrt{2\pi}$ and $(\pi - 2)/(4\pi)$. Solving $\int_0^m f(x)\,\mathrm{d}x = \frac{1}{2}$ numerically gives $m = 0.33724$. A simulation of $|Y|$ for 200 000 normal values with standard deviation $\frac{1}{2}$ gave mean 0.399 and variance 0.091, with 50.0% of values below 0.337. Sense check: the median 0.337 is below the mean 0.399, as it should be for a distribution with a long right tail.

Common slips

- Taking $\sigma^2 = 2$ or $\sigma = \frac{1}{4}$ from the exponent: match $2x^2$ with $x^2/(2\sigma^2)$, which gives $\sigma^2 = \frac{1}{4}$.
- Forgetting the factor 2: over half the line, the normal density only integrates to $\frac{1}{2}$.
- Solving $\Phi(2m) = \frac{1}{2}$, which gives $m = 0$. The half of X's probability below m corresponds to the middle half of Y's, so the condition is $\Phi(2m) = \frac{3}{4}$.

Beyond STEP

X is a *half-normal* random variable, and $|Y|$ for Y normal is the simplest example of a *transformation of a random variable*. In general, if $X = h(Y)$ then the density of X is obtained by summing the density of Y over all preimages, which is the factor 2 here. The same idea shows that Z^2 has a chi-squared distribution with one degree of freedom, and that is the start of a lot of statistics. Grimmett and Welsh, *Probability: An Introduction*, treats functions of random variables carefully.

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